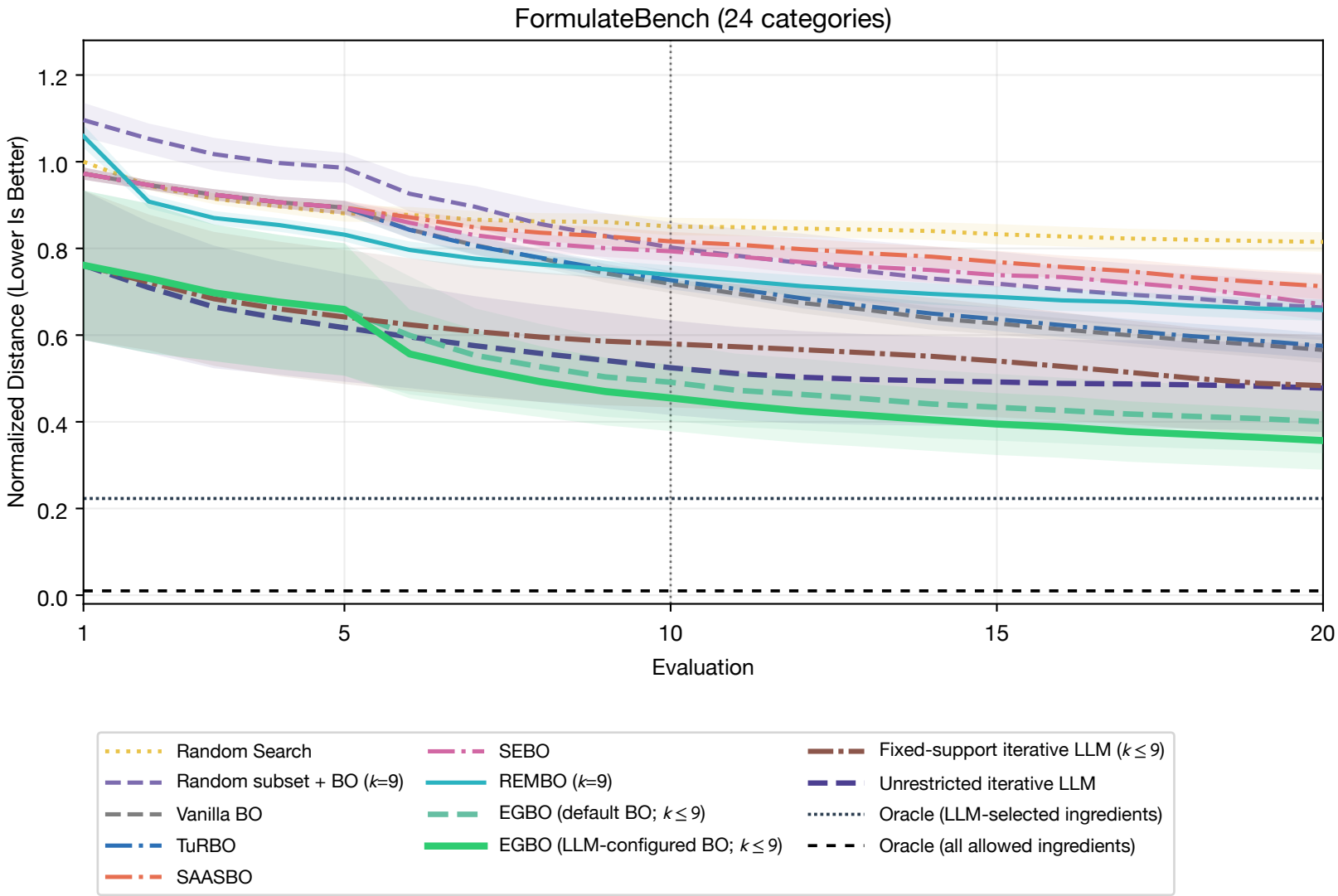
FormulateBench (24 categories)
Normalized Distance (Lower Is Better)
1.2
1.0
0.8
0.6
0.4
0.2
0.0
1
5
10
15
20
Evaluation
Random Search
Random subset + BO (k=9)
Vanilla BO
TuRBO
SAASBO
SEBO
REMBO (k=9)
EGBO (default BO; k ≤ 9)
EGBO (LLM-configured BO; k ≤ 9)
Fixed-support iterative LLM (k ≤ 9)
Unrestricted iterative LLM
Oracle (LLM-selected ingredients)
Oracle (all allowed ingredients)